

Wellbore Trajectory and Position Uncertainty

Python implementation and numerical verification

Julian Czettl | Independent engineering project

github.com/jczettl/wellbore-trajectory-uncertainty (code, full report and reproduction instructions)

I developed a Python workflow that reconstructs a three-dimensional well path from directional surveys and calculates the size and orientation of positional uncertainty using selected ISCWSA Revision 5 error sources.

Verification highlights: Volve reference coordinates reproduced within **8.7 mm** in each coordinate component; Using Balanced Tangential (BT) sensitivities, **10 of 11 selected covariance terms** matched three ISCWSA benchmarks to approximately numerical round-off. The remaining depth-stretch residual is quantified on page 2.

What I implemented

- **Survey processing and trajectory reconstruction:** checks for finite numerical inputs, increasing measured depth and inclination limits; Minimum Curvature Method (MCM) coordinates and dogleg severity from measured depth, inclination and azimuth.
- **Position-uncertainty propagation:** eleven selected depth, declination, sag and misalignment error terms. Random and correlated contributions determine uncertainty size and orientation, represented by a 3×3 North–East–Vertical covariance matrix at each survey station.
- **Analytical sensitivities:** BT and full MCM derivatives quantify how calculated position responds to measurement errors; both were checked against central finite differences.
- **Analysis and outputs:** reference comparisons, uncertainty ellipses, 3D visualization and breakdowns by error source to investigate the effects of well geometry.

Real-well application: Volve 15/9-F-11 A

The public Equinor survey contains **323 physical stations** to **3762 m measured depth**. The reconstructed end point is at approximately 3126.5 m true vertical depth (TVD).

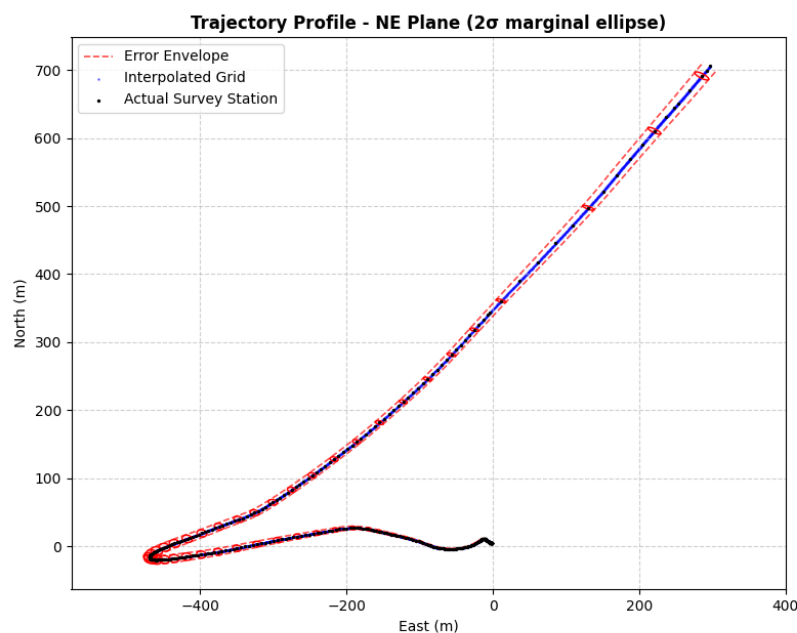


Figure 1: Volve 15/9-F-11 A: reconstructed top down view trajectory and selected 2σ marginal uncertainty ellipses using BT sensitivities.

Illustrative end-of-well uncertainty: horizontal ellipse semi-axes (radii) of approximately **11.0 m** and **3.52 m**; vertical uncertainty ± 11.62 m. All use a 2σ axis multiplier and describe the illustrative model output.

Numerical verification

Verification used supplied coordinates, three ISCWSA benchmark workbooks and numerical derivative checks.

Check	Result
Volve coordinates	Maximum absolute difference: 8.7 mm across North, East and TVD components. Supplied reference coordinates are rounded to 0.01 m.
ISCWSA trajectories	Maximum absolute TVD differences range from 4.9 to 35.9 mm across the three benchmark workbooks.
Selected covariance terms	With BT sensitivities, 10 of 11 terms agree to approximately numerical round-off. The maximum covariance-element residual for the remaining depth-stretch term (DSTG) is 9.3×10^{-5} m ² .
Analytical derivatives	Analytical BT and MCM derivatives agreed with central finite differences over the tested intervals; step-size convergence was checked.

DSTG depends on reconstructed TVD. Its residual is consistent with the observed TVD differences, although this does not uniquely establish the cause.

These tests assess numerical implementation, not physical survey accuracy.

BT versus MCM sensitivities

ISCWSA uses BT position sensitivities for uncertainty propagation along MCM trajectories. I implemented full MCM derivatives to test the effect of this approximation.

The comparison keeps the **MCM trajectory and eleven-term error model fixed**. Only sensitivities change; values below are maximum absolute differences over all Volve survey stations.

Uncertainty measure (2σ)	Maximum absolute MCM–BT difference
Horizontal ellipse semi-major axis	1.56 mm
Horizontal ellipse semi-minor axis	0.92 mm
Vertical uncertainty half-width	0.39 mm

Practical scale: the maximum **1.56 mm** change occurs at the end of the well, where the 2σ semi-major axis measures approximately **11 m**. This small effect is specific to the evaluated trajectory and model subset.

Scope and sources

The eleven-term educational model uses generic benchmark magnitudes, rather than the actual Volve survey-tool model. A complete tool-specific model, surface/tie-on uncertainty and multi-well anti-collision calculations are outside scope. The 2σ axis multiplier does not denote a 95% joint confidence region.

Reference data: ISCWSA Revision 5 benchmark workbooks; Equinor Volve dataset.